



Cambridge International AS & A Level

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COMPUTER SCIENCE**9618/31**

Paper 3 Advanced Theory

May/June 2025**1 hour 30 minutes**

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].
- No marks will be awarded for using brand names of software packages or hardware.

This document has **16** pages. Any blank pages are indicated.

- 1 A programmer is writing a program to manage bookings for a small taxi company. The programmer requires some user-defined data types.

- (a) Write a **pseudocode** statement to declare the enumerated data type, `Vehicle`, to hold the identity code of each of the company's taxis:

M100, M230, T101, T102, T120, T150

.....
 [2]

- (b) Write **pseudocode** statements to declare the composite data type, `Booking`, to hold data about taxi bookings. The data required includes:

- booking number (any combination of letters and numbers)
- destination
- client name
- client telephone number
- date of departure
- address for pick-up
- the identity code of the taxi used.

Use the most appropriate data type in each case, including the enumerated data type from part (a).

.....

 [4]





2 Numbers are stored in a computer using binary floating-point representation with:

- 10 bits for the mantissa
- 6 bits for the exponent
- two's complement form for both the mantissa and the exponent.

(a) Write the normalised floating-point representation of the following binary number using this system.

0.00000011010111

Mantissa

--	--	--	--	--	--	--	--	--	--

Exponent

--	--	--	--	--	--

[2]

(b) Calculate the normalised binary floating-point representation of -25.3125 in this system. Show your working.

Mantissa

--	--	--	--	--	--	--	--	--	--

Exponent

--	--	--	--	--	--

Working

.....

.....

.....

.....

.....

.....

.....

[4]





- 3 (a) The Application Layer and Transport Layer are two layers of the TCP/IP protocol suite.

Describe the purpose of the Application Layer **and** the purpose of the Transport Layer.

Purpose of Application Layer

.....

.....

.....

.....

Purpose of Transport Layer

.....

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.....

.....

[5]

- (b) Describe packet switching as a method of transmitting messages across the internet.

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.....

.....

.....

.....

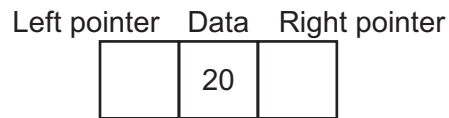
.....

.....

..... [4]



- 4 (a) A linked list of nodes is used to store an ordered list of integers. Each node consists of the data, a left pointer and a right pointer, for example:

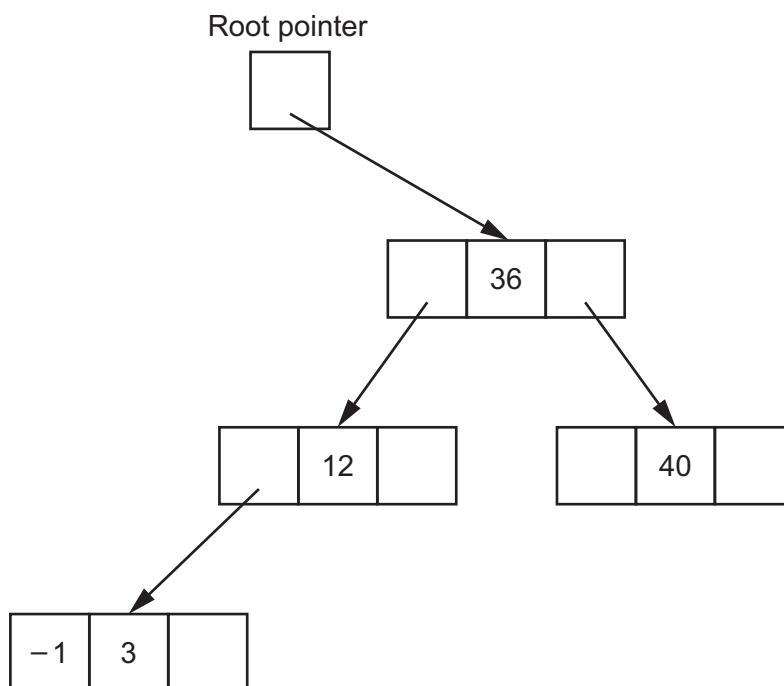


The linked list will be organised as a binary tree.

-1 is used to represent a null pointer.

Complete the binary tree, including null pointers, to show how the data will be organised after the following integers have been added:

6, 15, 41, 66



[4]

- (b) Describe what is meant by recursion.

.....

.....

.....

..... [2]





- (c) A binary tree is a suitable Abstract Data Type (ADT) that a designer can implement using recursive algorithms.

Identify **one other** ADT that a designer can implement using recursive algorithms.

..... [1]

- 5 This truth table represents a logic circuit.

INPUT				OUTPUT
A	B	C	D	Z
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

- (a) Write the Boolean logic expression that corresponds to the given truth table as the sum-of-products.

Z =

..... [2]





(b) (i) Complete the Karnaugh map (K-map) for the given truth table.

		AB			
		00	01	11	10
CD	00				
	01				
	11				
	10				

[2]

- (ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]
- (iii) Write the Boolean logic expression from your answer to part (b)(ii) as the simplified sum-of-products.

$Z =$

..... [2]

6 Describe the process of executing a program using an interpreter.

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.....

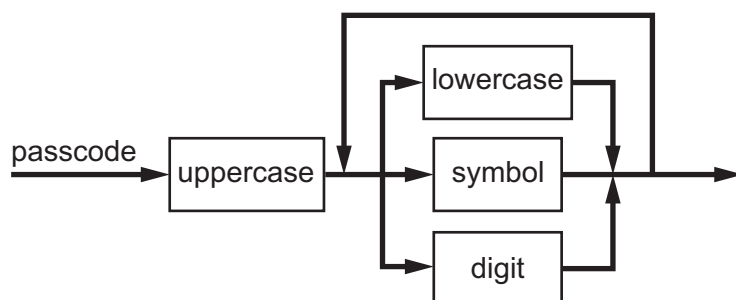
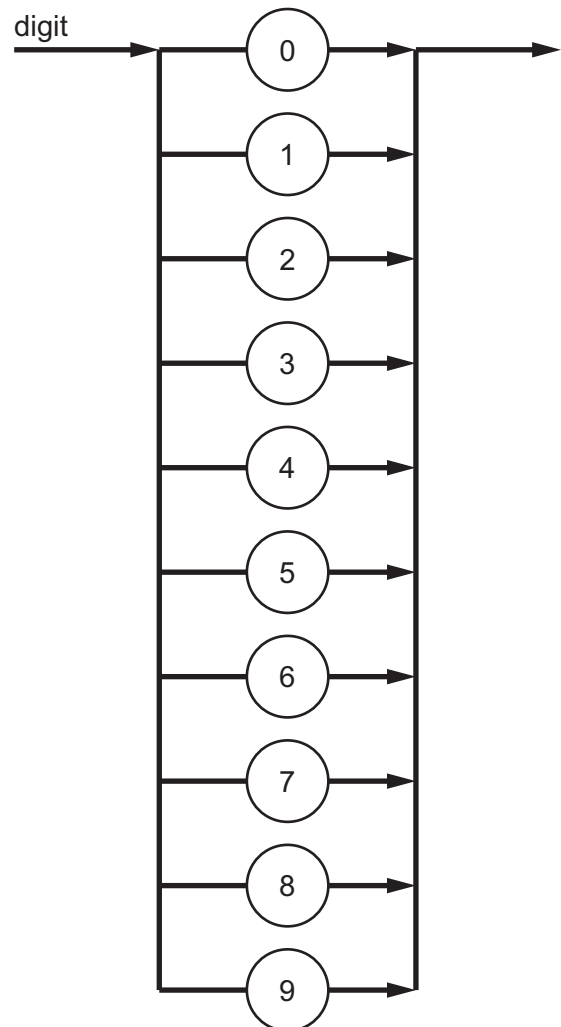
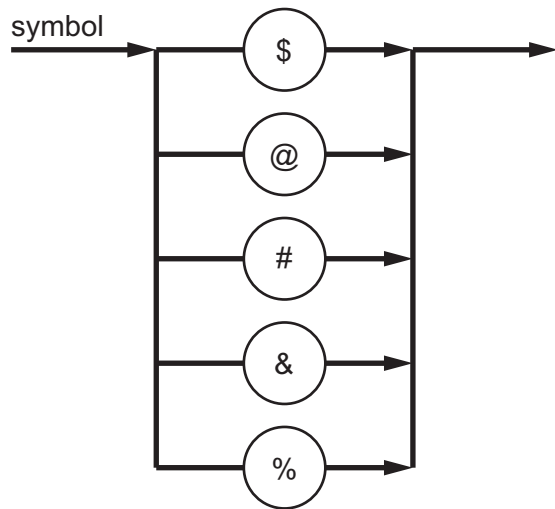
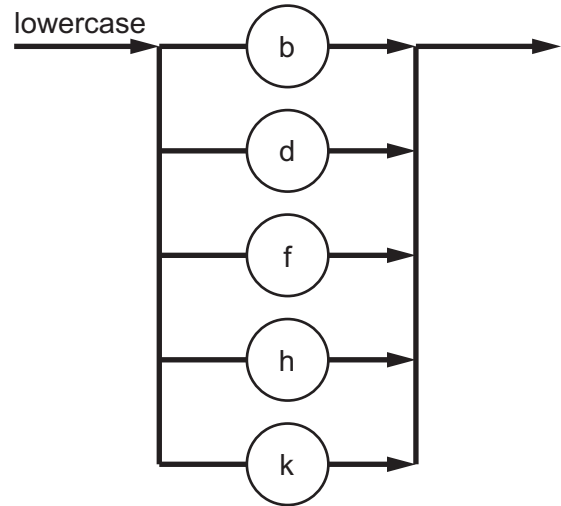
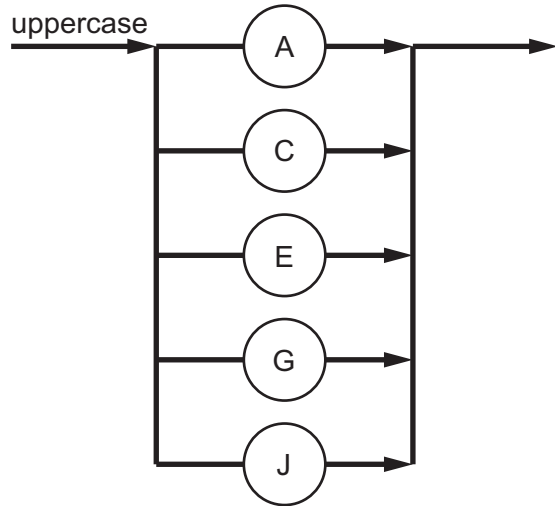
.....

..... [4]





7 Several syntax diagrams are shown.





(a) State why each **passcode** is invalid for the given syntax diagrams.

#Jd7

Reason

.....

C%6A

Reason

.....

[2]

(b) Complete the Backus-Naur Form (BNF) for <uppercase> and <passcode>.

<uppercase> ::=

.....

<passcode> ::=

.....

.....

.....

.....

.....

[4]





8 (a) Describe what is meant by **multi-tasking** and how it benefits process management.

.....

.....

.....

.....

..... [2]

(b) Explain the function of the shortest remaining time scheduling routine **and** give a benefit of this routine.

Function

.....

.....

.....

.....

.....

.....

Benefit

..... [4]

9 Secure Socket Layer (SSL) and Transport Layer Security (TLS) are two protocols.

(a) State **two** functions of SSL/TLS.

1

.....

2

..... [2]

(b) Give **two** examples of situations where the use of SSL/TLS would be appropriate.

1

.....

2

..... [2]





10 (a) Describe the purpose of a graph when used in an Artificial Intelligence (AI) system.

.....

.....

.....

..... [2]

(b) Explain the use of artificial neural networks in Deep Learning.

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..... [4]



- 11

Attribute	Data type	Description
DateSeen	DATE	date of treatment
Treatments	STRING	treatments given
Medications	STRING	medications prescribed

Doctors are identified by their name, for example, A N Other.

Complete the class diagram for Appointment, to include:

- attribute and data type for the identification of the patient
- attribute and data type for the identification of the doctor
- methods to assign date seen, treatments given and medications prescribed
- method to return the date seen and the attributes for the patient and the doctor.

Appointment	
DateSeen	: DATE
.....	:
.....	:
Treatments	: STRING
Medications	: STRING
 SetPatientID(PatientNumber : INTEGER) SetDoctor(DoctorID : STRING) GetTreatments() GetMedications()	



- (b) (i) Identify the object-oriented programming (OOP) feature whose function includes restricting external access to the data.

..... [1]

- (ii) Describe what is meant by the OOP feature **inheritance**.

.....

 [2]

- 12 The pseudocode algorithm checks whether a location in a stock file `StockList.dat` is empty or not. The location is given by the user. If the location is empty, a suitable message is displayed, otherwise the item stored at that location is displayed.

Complete this file-handling pseudocode algorithm.

```
DECLARE Location : INTEGER
DECLARE Item : STRING
DECLARE Continue : BOOLEAN
DECLARE Answer : CHAR
Continue ← TRUE
```

```
OPENFILE .....
WHILE Continue
  OUTPUT "Enter a location between 1 and 500: "
  INPUT Location
  .....
```

```
  GETRECORD .....
  IF Item = "" THEN
    OUTPUT "This record is missing."
  ELSE
    OUTPUT "The item in stock is ", .....
  ENDIF
  OUTPUT "Another location (Y or N)?"
  INPUT Answer
  IF Answer <> 'Y' THEN
    Continue ← FALSE
  ENDIF
ENDWHILE
```

```
.....
OUTPUT "End of program"
```

[5]









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